
Module 1: Two Pointers

1. Two Sum in Sorted Array (Easy)

Problem

Given a sorted array and a target value, return the 1-based indices of two numbers whose sum equals the target.

Constraints

- $2 \leq n \leq 10^5$
- $-10^9 \leq \text{arr}[i] \leq 10^9$

Edge Cases

- Negative values
- Duplicates
- Answer always exists

Expected

- Time: $O(n)$
 - Space: $O(1)$
-

2. Container With Most Water (Medium)

Problem

Given heights of vertical lines, find maximum water container.

Constraints

- $n \leq 100000$

Edge Cases

- All same heights
- Strictly increasing
- Strictly decreasing

Expected

- $O(n)$
-

3. Trapping Rain Water (Hard)

Problem

Calculate trapped rain water.

Example

Input:

[4,2,0,3,2,5]

Output:

9

Expected

- $O(n)$
 - $O(1)$
-

Module 2: Sliding Window

4. Maximum Sum Subarray of Size K (Easy)

Problem

Find maximum sum of any subarray of size K.

Constraints

- $n \leq 10^5$

Edge Cases

- $K = 1$
- $K = n$

Expected

- $O(n)$
-

5. Longest Substring Without Repeating Characters (Medium)

Problem

Return length of longest substring containing unique characters.

Example

abcabcbb -> 3

Expected

- $O(n)$
-

6. Minimum Window Substring (Hard)

Problem

Find smallest window containing all characters of target string.

Example

S = ADOBECODEBANC

T = ABC

Output = BANC

Expected

- $O(n)$
-

Module 3: Kadane's Algorithm

7. Maximum Subarray Sum (Easy)

Problem

Find largest contiguous subarray sum.

Constraints

$-10^4 \leq \text{nums}[i] \leq 10^4$

Edge Cases

[-5,-2,-7]

Expected

- $O(n)$
-

8. Maximum Circular Subarray Sum (Medium)

Problem

Subarray may wrap around end of array.

Example

[5,-3,5]

Output: 10

Expected

- $O(n)$
-

9. Maximum Product Subarray (Hard)

Problem

Find contiguous subarray having largest product.

Edge Cases

[-2]

[-2,0,-1]

Expected

- $O(n)$
-

Module 4: Prefix Sum

10. Range Sum Query (Easy)

Problem

Answer q range sum queries.

Expected

- Preprocessing $O(n)$
 - Query $O(1)$
-

11. Subarray Sum Equals K (Medium)

Problem

Count subarrays whose sum equals k.

Example

nums=[1,1,1]

k=2

Output=2

Expected

- $O(n)$
-

12. Longest Subarray With Sum K (Hard)

Problem

Return length of longest subarray having sum exactly K.

Constraints

Negative numbers allowed

Expected

- $O(n)$
-

Module 5: Merge / Union / Intersection

13. Merge Sorted Arrays (Easy)

Problem

Merge two sorted arrays.

Expected

- $O(n+m)$
-

14. Union of Two Arrays (Medium)

Problem

Return distinct union of two arrays.

Example

A=[1,2,3]

B=[2,3,4]

Output=[1,2,3,4]

Expected

- $O(n+m)$
-

15. Intersection of Two Arrays II (Hard)

Problem

Intersection should include duplicates.

Example

A=[1,2,2,1]

B=[2,2]

Output=[2,2]

Expected

- $O(n+m)$
-

Module 6: In-place Rearrangement

16. Sort Colors (Dutch National Flag) (Easy)

Problem

Sort array containing only 0,1,2.

Expected

- $O(n)$
 - $O(1)$
-

17. Missing Number Using Cyclic Sort (Medium)

Problem

Array contains numbers from 0 to n with one missing.

Example

[3,0,1]

Output:

2

Expected

- $O(n)$
- $O(1)$

18. First Missing Positive (Hard)

Problem

Find smallest positive integer missing from array.

Example

[3,4,-1,1]

Output:

2

Constraints

$O(n)$ time

$O(1)$ space

Edge Cases

[]

[-1,-2]

[1,2,3]
